

DXF Tags

A Drawing Interchange File is simply an ASCII text file with a file type of .dxf and special formatted text. The basic file structure are DXF tags, a DXF tag consist of a DXF group code as an integer value on its own line and a the DXF value on the following line. In the ezdxf documentation DXF tags will be written as (group code, value).

With the introduction of extended symbol names in DXF R2000, the 255-character limit for strings has been increased to 2049 single-byte characters not including the newline at the end of the line. Nonetheless its safer to use only strings with 255 and less characters, because its not clear if this fact is true for ALL string group codes or only for symbols like layer- or text style names and not all 3rd party libraries may handle this fact correct. The MTEXT content and binary data is still divided into chunks with less than 255 characters.

Group codes are indicating the value type:

Group Code	Value Type
0-9	String
10-39	Double precision 3D point value
40-59	Double-precision floating-point value
60-79	16-bit integer value
90-99	32-bit integer value
100	String
102	String
105	String representing hexadecimal (hex) handle value
110-119	Double precision floating-point value
120-129	Double precision floating-point value

Group Code	Value Type
130-139	Double precision floating-point value
140-149	Double precision scalar floating-point value
160-169	64-bit integer value
170-179	16-bit integer value
210-239	Double-precision floating-point value
270-279	16-bit integer value
280-289	16-bit integer value
290-299	Boolean flag value
300-309	Arbitrary text string
310-319	String representing hex value of binary chunk
320-329	Arbitrary pointer, hex object ID, not translated during INSERT and XREF operations
330-339	Soft-pointer, hex object ID, translated during INSERT and XREF operations
340-349	Hard-pointer, hex object ID, translated during INSERT and XREF operations
350-359	Soft-owner, hex object ID, translated during INSERT and XREF operations
360-369	Hard-owner, hex object ID, translated during INSERT and XREF operations
370-379	16-bit integer value
380-389	16-bit integer value
390-399	String representing hex handle value
400-409	16-bit integer value
410-419	String
420-429	32-bit integer value
430-439	String
440-449	32-bit integer value
450-459	Long

Group Code	Value Type
460-469	Double-precision floating-point value
470-479	String
480-481	Hard-pointer, hex object ID, translated during INSERT and XREF operations
999	Comment (string)
1000-1009	String
1010-1059	Double-precision floating-point value
1060-1070	16-bit integer value
1071	32-bit integer value

Explanation for some important group codes:

Group Code	Meaning
0	DXF structure tag, entity start/end or table entries
1	The primary text value for an entity
2	A name: Attribute tag, Block name, and so on. Also used to identify a DXF section or table name.
3-4	Other textual or name values
5	Entity handle as hex string (fixed)
6	Line type name (fixed)
7	Text style name (fixed)
8	Layer name (fixed)
9	Variable name identifier (used only in HEADER section of the DXF file)
10	Primary X coordinate (start point of a Line or Text entity, center of a Circle, etc.)
11-18	Other X coordinates
20	Primary Y coordinate. 2n values always correspond to 1n values and immediately follow them in the file (expected by ezdxf!)
21-28	Other Y coordinates

Group Code	Meaning
30	Primary Z coordinate. 3n values always correspond to 1n and 2n values and immediately follow them in the file (expected by ezdxf!)
31-38	Other Z coordinates
39	This entity's thickness if nonzero (fixed)
40-48	Float values (text height, scale factors, etc.)
49	Repeated value - multiple 49 groups may appear in one entity for variable length tables (such as the dash lengths in the LTYPE table). A 7x group always appears before the first 49 group to specify the table length
50-58	Angles in degree
62	Color number (fixed)
66	"Entities follow" flag (fixed), only in INSERT and POLYLINE entities
67	Identifies whether entity is in modelspace (0) or paperspace (1)
68	Identifies whether viewport is on but fully off screen, is not active, or is off
69	Viewport identification number
70-78	Integer values such as repeat counts, flag bits, or modes
105	DIMSTYLE entity handle as hex string (fixed)
210, 220, 230	X, Y, and Z components of extrusion direction (fixed)
310	Proxy entity graphics as binary encoded data
330	Owner handle as hex string
347	MATERIAL handle as hex string
348	VISUALSTYLE handle as hex string
370	Lineweight in mm times 100 (e.g. 0.13mm = 13).
390	PLOTSTYLE handle as hex string
420	True color value as 0x00RRGGBB 24-bit value
430	Color name as string
440	Transparency value 0x020000TT 0 = fully transparent / ---
999	Comments

Extended Data

DXF R2018 Reference

Extended data (XDATA) is created by AutoLISP or ObjectARX applications but any other application like *ezdxf* can also define XDATA. If an entity contains extended data, it **follows** the entity’s normal definition.

But extended group codes (≥ 1000) can appear **before** the XDATA section, an example is the BLOCKBASEPOINTPARAMETER entity in AutoCAD Civil 3D or AutoCAD Map 3D.

Group Code	Description
1000	Strings in extended data can be up to 255 bytes long (with the 256th byte reserved for the null character)
1001	(fixed) Registered application name (ASCII string up to 31 bytes long) for XDATA
1002	(fixed) An extended data control string can be either '{' or '}'. These braces enable applications to organize their data by subdividing the data into lists. Lists can be nested.
1003	Name of the layer associated with the extended data
1004	Binary data is organized into variable-length chunks. The maximum length of each chunk is 127 bytes. In ASCII DXF files, binary data is represented as a string of hexadecimal digits, two per binary byte
1005	Database Handle of entities in the drawing database, see also: About 1005 Group Codes
1010, 1020, 1030	Three real values, in the order X, Y, Z. They can be used as a point or vector record that will not be modified at any transformation of the entity.
1011, 1021, 1031	a WCS point that is moved, scaled, rotated and mirrored along with the entity
1012, 1012, 1022	a WCS displacement that is scaled, rotated and mirrored along with the entity, but is not moved
1013, 1023, 1033	a WCS direction that is rotated and mirrored along with the entity, but is not moved or scaled

Group Code	Description
1040	A real value
1041	Distance, a real value that is scaled along with the parent entity
1042	Scale Factor, also a real value that is scaled along with the parent. The difference between a distance and a scale factor is application-defined
1070	A 16-bit integer (signed or unsigned)
1071	A 32-bit signed (long) integer

The (1001, ...) tag indicates the beginning of extended data. In contrast to normal entity data, with extended data the same group code can appear multiple times, and **order is important**.

Extended data is grouped by registered application name. Each registered application group begins with a (1001, APPID) tag, with the application name as APPID string value. Registered application names correspond to APPID symbol table entries.

An application can use as many APPID names as needed. APPID names are permanent, although they can be purged if they aren't currently used in the drawing. Each APPID name can have **no more than one data group** attached to each entity. Within an application group, the sequence of extended data groups and their meaning is defined by the application.

String value encoding

String values stored in a DXF file is plain ASCII or UTF-8, AutoCAD also supports CIF (Common Interchange Format) and MIF (Maker Interchange Format) encoding. The UTF-8 format is only supported in DXF R2007 and later.

Ezdxflib on import converts all strings into Python unicode strings without encoding or decoding CIF/MIF.

String values containing Unicode characters are represented with control character sequences `\U+nnnn` . (e.g. `r'TEST\U+7F3A\U+4E4F\U+89E3\U+91CA\U+6B63THIS\U+56FE '`)

To support the DXF unicode encoding ezdxflib registers an encoding code  `stable`  `dxflib.backslash_replace`, defined in `ezdxflib.dxf.encoding()` .

String values can be stored with these dxf group codes:

- 0 - 9
- 100 - 101
- 300 - 309
- 410 - 419
- 430 - 439
- 470 - 479
- 999 - 1003

Multi tag text (MTEXT)

If the text string is less than 250 characters, all characters appear in tag (1, ...). If the text string is longer than 250 characters, the string is divided into 250-character chunks, which appear in one or more (3, ...) tags. If (3, ...) tags are used, the last group is a (1, ...) tag and has fewer than 250 characters:

```
3
... TwoHundredAndFifty Characters ....
3
... TwoHundredAndFifty Characters ....
1
less than TwoHundredAndFifty Characters
```

As far I know this is only supported by the MTEXT entity.

 See also

[DXF File Encoding](#)

DXF R13 and later tag structure

With the introduction of DXF R13 Autodesk added additional group codes and DXF tag structures to the DXF Standard.

Subclass Markers

Subclass markers (100, Subclass Name) divides DXF objects into several sections. Group codes can be reused in different sections. A subclass ends with the following subclass marker or at the beginning of xdata or the end of the object. See Subclass Marker Example in the DXF Reference.

Quote about group codes from the DXF reference

Some group codes that define an entity always appear; others are optional and appear only if their values differ from the defaults.

Do not write programs that **rely on the order given here**. The end of an entity is indicated by the next 0 group, which begins the next entity or indicates the end of the section.

Note: Accommodating DXF files from future releases of AutoCAD will be easier if you write your DXF processing program in a table-driven way, ignore undefined group codes, and make no assumptions about the order of group codes in an entity. With each new AutoCAD release, new group codes will be added to entities to accommodate additional features.

Usage of group codes in subclasses twice

Some later entities contains the same group code twice for different purposes, so order in the sense of which one comes first is important. (e.g. ATTDEF group code 280)

Tag order is sometimes important especially for AutoCAD

In LWPOLYLINE the order of tags is important, if the *count* tag is not the first tag in the AcDbPolyline subclass, AutoCAD will not close the polyline when the *close* flag is set, by the way other applications like BricsCAD ignores the tag order and renders the polyline always correct.

Extension Dictionary

The extension dictionary is an optional sequence that stores the handle of a DICTONARY object that belongs to the current object, which in turn may contain entries. This facility allows attachment of arbitrary database objects to any database object. Any object or entity may have this section.

The extension dictionary tag sequence:

```
102
{ACAD_XDICTIONARY
360
Hard-owner ID/handle to owner dictionary
102
}
```

Persistent Reactors

Persistent reactors are an optional sequence that stores object handles of objects registering themselves as reactors on the current object. Any object or entity may have this section.

The persistent reactors tag sequence:

```
102
{ACAD_REACTORS
330
first Soft-pointer ID/handle to owner dictionary
330
second Soft-pointer ID/handle to owner dictionary
...
102
}
```

Application-Defined Codes

Starting at DXF R13, DXF objects can contain application-defined codes (XDATA). This application-defined codes can contain any tag except (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 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1008, 1009, 1010, 1011, 1012, 1013, 1014, 1015, 1016, 1017, 1018, 1019, 1020, 1021, 1022, 1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1031, 1032, 1033, 1034, 1035, 1036, 1037, 1038, 1039, 1040, 1041, 1042, 1043, 1044, 1045, 1046, 1047, 1048, 1049, 1050, 1051, 1052, 1053, 1054, 1055, 1056, 1057, 1058, 1059, 1060, 1061, 1062, 1063, 1064, 1065, 1066, 1067, 1068, 1069, 1070, 1071, 1072, 1073, 1074, 1075, 1076, 1077, 1078, 1079, 1080, 1081, 1082, 1083, 1084, 1085, 1086, 1087, 1088, 1089, 1090, 1091, 1092, 1093, 1094, 1095, 1096, 1097, 1098, 1099, 1100, 1101, 1102, 1103, 1104, 1105, 1106, 1107, 1108, 1109, 1110, 1111, 1112, 1113, 1114, 1115, 1116, 1117, 1118, 1119, 1120, 1121, 1122, 1123, 1124, 1125, 1126, 1127, 1128, 1129, 1130, 1131, 1132, 1133, 1134, 1135, 1136, 1137, 1138, 1139, 1140, 1141, 1142, 1143, 1144, 1145, 1146, 1147, 1148, 1149, 1150, 1151, 1152, 1153, 1154, 1155, 1156, 1157, 1158, 1159, 1160, 1161, 1162, 1163, 1164, 1165, 1166, 1167, 1168, 1169, 1170, 1171, 1172, 1173, 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1340, 1341, 1342, 1343, 1344, 1345, 1346, 1347, 1348, 1349, 1350, 1351, 1352, 1353, 1354, 1355, 1356, 1357, 1358, 1359, 1360, 1361, 1362, 1363, 1364, 1365, 1366, 1367, 1368, 1369, 1370, 1371, 1372, 1373, 1374, 1375, 1376, 1377, 1378, 1379, 1380, 1381, 1382, 1383, 1384, 1385, 1386, 1387, 1388, 1389, 1390, 1391, 1392, 1393, 1394, 1395, 1396, 1397, 1398, 1399, 1400, 1401, 1402, 1403, 1404, 1405, 1406, 1407, 1408, 1409, 1410, 1411, 1412, 1413, 1414, 1415, 1416, 1417, 1418, 1419, 1420, 1421, 1422, 1423, 1424, 1425, 1426, 1427, 1428, 1429, 1430, 1431, 1432, 1433, 1434, 1435, 1436, 1437, 1438, 1439, 1440, 1441, 1442, 1443, 1444, 1445, 1446, 1447, 1448, 1449, 1450, 1451, 1452, 1453, 1454, 1455, 1456, 1457, 1458, 1459, 1460, 1461, 1462, 1463, 1464, 1465, 1466, 1467, 1468, 1469, 1470, 1471, 1472, 1473, 1474, 1475, 1476, 1477, 1478, 1479, 1480, 1481, 1482, 1483, 1484, 1485, 1486, 1487, 1488, 1489, 1490, 1491, 1492, 1493, 1494, 1495, 1496, 1497, 1498, 1499, 1500, 1501, 1502, 1503, 1504, 1505, 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```
102
{YOURAPPID
...
102
}
```

`(102, 'YOURAPPID}')` is also a valid closing tag:

```
102
{YOURAPPID
...
102
YOURAPPID}
```

All groups defined with a beginning (102, ...) appear in the DXF reference before the first subclass marker, I don't know if these groups can appear after the first or any subclass marker. Ezdxf accepts them at any position, and by default ezdxf adds new app data in front of the first subclass marker to the first tag section of an DXF object.

Exception XRECORD: Tags with group code 102 and a value string without a preceding "{" or the scheme "YOURAPPID}", should be treated as usual group codes.

Embedded Objects

The concept of embedded objects was introduced with AutoCAD 2018 (DXF version AC1032) and this is the only information I found about it at the Autodesk knowledge base: Embedded and Encapsulated Objects

Quote from Embedded and Encapsulated Objects:

For DXF filing, the embedded object must be filed out and in after all the data of the encapsulating object has been filed out and in.

A separator is needed between the encapsulating object's data and the subsequent embedded object's data. The separator must be similar in function to the group 0 or 100 in that it must cause the filer to stop reading data. The normal DXF group code 0 cannot be used because DXF proxies use it to determine when to stop reading data. The group code 100 could have been used, but it might have caused confusion when manually reading a DXF file, and there was a need to distinguish when an embedded object is about to be written out in order to do some internal bookkeeping. Therefore, the DXF group code 101 was introduced.

Hard facts:

- Only used in ATTRIB, ATTDEF (embedded MTEXT) and MTEXT (columns) in DXF R2018.
- Embedded object start with (101, "Embedded Object") tag
- Embedded object is appended to the encapsulated object
- Embedded object tags can contain any group code except the DXF structure tag (0, ...)

Unconfirmed assumptions:

- The embedded object is written before the Extended Data. No examples for entities including embedded objects and XDATA at the same time.
- XDATA sections replaced by embedded objects, at least for the MTEXT entity
- The encapsulating object can contain more than one embedded object.
- Embedded objects separated by (101, "Embedded Object") tags
- every entity can contain embedded objects

Real world example from an AutoCAD 2018 file:

100 <<< start of encapsulating object

AcDbMText

10

2762.148

20

2327.073

30

0.0

40

2.5

41

18.852

46

0.0

71

1

72

5

1

{\fArial|b0|i0|c162|p34;CHANGE;\P\P\PTXT}

73

1

44

1.0

101 <<< start of embedded object

Embedded Object

70

1

10

1.0

20

0.0

30

0.0

11

2762.148

21

2327.073

31

0.0

40

18.852

41

0.0

42

15.428

43

15.043

71

2

72

1

44

18.852

45
12.5
73
0
74
0
46
0.0